

Sustainable energy transition toward a zero-carbon future in major Global South economies: Roles of green innovation, institutions, and geopolitical risk[☆]

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ABSTRACT

This study examines the renewable energy transition in five major Global South economies, namely Brazil, China, India, Indonesia, and Thailand, by focusing on the roles of green innovation, institutional quality, and geopolitical risk. Anchored in Sustainable Development Goal 7 (SDG-7), the study moves beyond conventional determinant-based approaches by framing energy transition as a sustainability governance process shaped by technological capacity, institutional credibility, and external geopolitical uncertainty. Using a balanced panel dataset for the five countries over 2002–2023, the analysis evaluates long-run associations between green innovation, regulatory quality, geopolitical risk, and renewable energy outcomes. The findings show that green innovation supports renewable energy transition, while institutional quality strengthens this process by improving policy credibility, investment conditions, and regulatory stability. By contrast, geopolitical risk constrains renewable energy deployment by increasing uncertainty and weakening long-term investment incentives. The results further suggest that progress toward SDG-7 in major emerging economies depends not only on technological advancement but also on institutions' ability to sustain innovation-oriented energy policies amid geopolitical instability. By integrating geopolitical risk into the innovation-institution nexus, this research offers a more context-sensitive explanation of renewable energy transition in emerging economies and provides policy-relevant insights for designing resilient SDG-7 strategies.

Introduction

Progress toward Sustainable Development Goal 7 (SDG-7) has become a defining challenge of sustainability governance in emerging economies. Advances in renewable energy (REN) technologies, storage systems, and digital infrastructure have enhanced the technical feasibility of cleaner energy systems, but achieving SDG-7 ultimately depends on governance structures that sustain policy consistency and long-term investment. However, the diffusion and effective adoption of these technologies remain highly uneven across countries, particularly in major emerging economies such as Brazil, China, India, Indonesia, and

Thailand, where institutional capacity, infrastructure gaps, and exposure to external shocks constrain long-term technological adaptation. Geopolitical uncertainty emerges here as a structural constraint that may weaken institutional credibility, distort policy priorities, and delay REN investments. Thus, understanding progress toward SDG-7 requires examining how governance quality conditions the effectiveness of green innovation (GRI) under geopolitical instability.

Within this broader governance context, clean energy transition (ENT) represents a core pathway for advancing affordable, reliable, and low-carbon energy systems. As the effects of climate change worsen worldwide, decarbonizing energy systems has become not only an

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environmental priority but also a key requirement for economic resilience, energy security, and social well-being (Lazaro, Usuriaga-Najera, Neto, Grimoni, & Jacobi, 2025). SDG-7 focuses on transforming global energy systems to provide access to affordable, reliable, sustainable, and modern energy for everyone. Specifically, sub-goal 7.2 seeks to increase the share of REN in the global energy mix by 2030, while sub-goal 7.a highlights the importance of international cooperation to advance research and technological innovation in REN (Anto, Mehta, & Murti, 2024; Erdas, Caglar, Uche, & Ahmed, 2025). Despite significant progress, global efforts still fall short of achieving the 1.5 °C target (Hartvig et al., 2025). According to the International Energy Agency (IEA), the worldwide share of REN increased from 7.24% in 2002 to 12.47% in 2021 (IEA, 2025). However, rising global energy demand continues to offset some of these improvements. Developing economies in the Global South face the core of this dual challenge: they need to meet growing energy demands driven by economic growth (EG), industrialization, and urbanization, while also reducing dependence on fossil fuels and lowering carbon emissions. Countries like Brazil, China, India, Indonesia, and Thailand exemplify this complex situation.

According to Our World in Data (OWD, 2025), Brazil consistently held the largest share of REN during the observed period, increasing from 38.8% in 2002 to about 48.9% in 2023. China's share of REN, around 6% in the early 2000s, steadily rose to 15.5% by 2023. Indonesia and Thailand, starting from much lower bases, showed gradual increases, especially over the last decade. In contrast, India's trend has been more volatile and slower. These patterns show apparent differences between countries: some have made significant progress, while ENT remains limited in others. Key factors influencing the trajectory and speed of ENT include regulatory quality (RQ), geopolitical risk (GEO), and their interactions with ENT dynamics. RQ measures governments' ability to develop and implement policies that promote private-sector development (Worldwide Governance Indicators (WGI), 2025). In

essence, RQ reflects the stability, predictability, and efficiency of a country's institutional framework for managing market functions (Abozeid, Elamer, & Attia, 2025). In the energy sector, higher RQ boosts investor confidence, minimizes bureaucratic hurdles, and encourages financial investment in REN projects (Odoro, Uzougbo, & Ugwu, 2024). According to the WGI (2025), RQ in the Global South generally ranges from low to moderate. China, India, and Indonesia mostly have negative or near-zero values, while Brazil and Thailand perform better but with volatility and occasional declines. This suggests that the institutional frameworks supporting ENT are unstable in many of these countries.

Another important factor shaping ENT is GEO. Countries in the Global South tend to be more vulnerable to geopolitical shifts due to their reliance on energy imports, exposure to regional conflict, and sensitivity to foreign policy changes. As asserted by Caldara and Iacoviello (2022), GEO encompasses political instability, military tensions, international sanctions, dependence on imported energy, and diplomatic uncertainty. The Russia-Ukraine conflicts since 2014, escalating into full-scale war in 2022, have notably increased global geopolitical uncertainty. These tensions have affected energy markets, driven up fossil fuel prices, and reshaped global trade, impacting energy security and transition strategies in the Global South. The Russia-Ukraine conflict has shown how geopolitical instability can affect the strength of institutions and policies in developing economies, influencing ENT.

Fig. 1 displays GEO data for the Global South, showing an upward trend after major conflicts, including the Russia-Ukraine war. GEO increased considerably across the five Global South economies following major crises. Although exposure levels vary across countries, the overall pattern suggests that geopolitical instability has become an increasingly important structural factor shaping energy security, investment expectations, and REN dynamics in these nations. At the same time, recent evidence suggests that geopolitical tensions may have mixed effects,

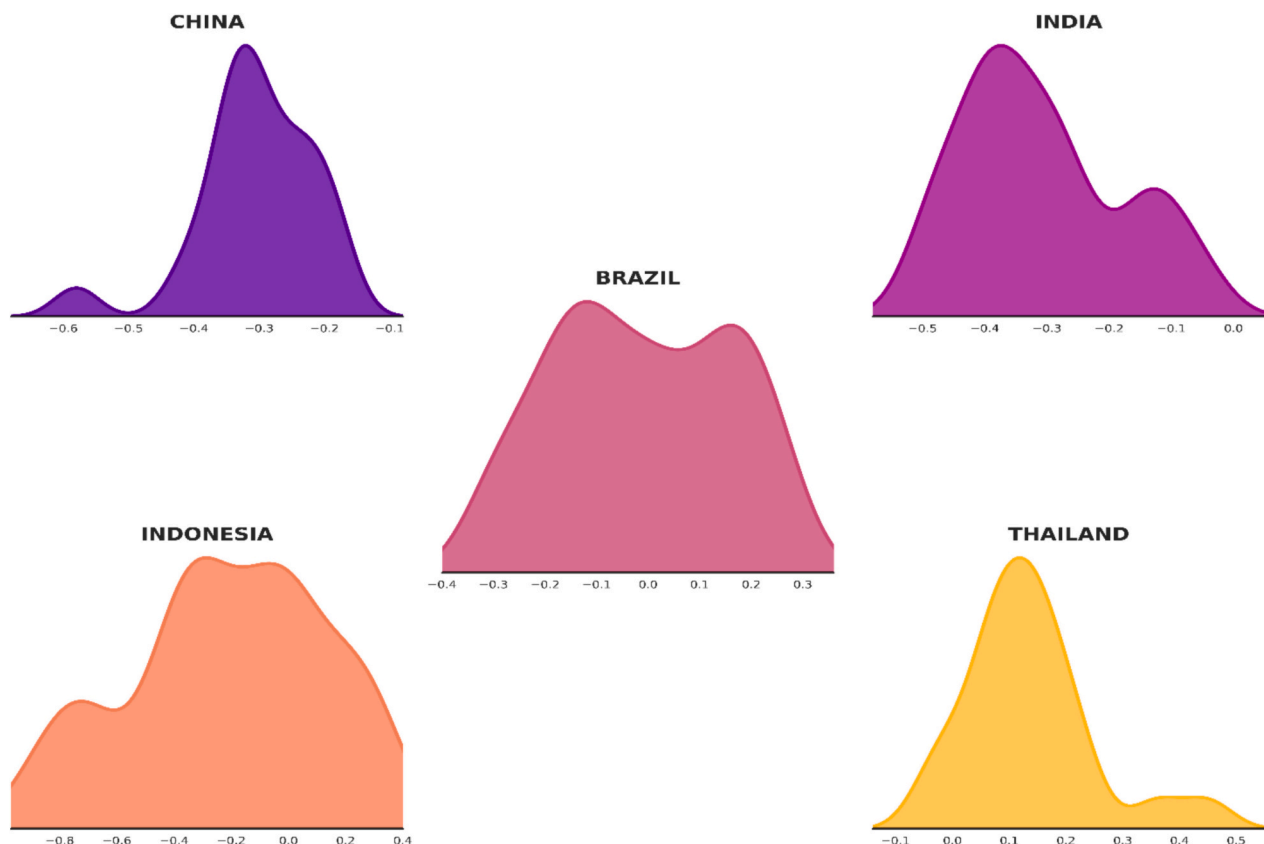


Fig. 1. GEO Data for Global South Countries (Brazil, China, India, Indonesia, Thailand)
Source: Caldara and Iacoviello (2022).

including increased uncertainty, while also strengthening incentives for energy diversification and energy security (He, Lee, & Sharma, 2025).

Finally, the technological side of ENT is shaped by GRI, enabling the development of environmentally friendly technologies (Cheng, Wang, Yang, & Li, 2024), improving energy efficiency (Dong, Zhao, & Taghizadeh-Hesary, 2023), and reducing carbon intensity, which directly affects the speed and success of ENT. Recent OECD (2025) data shows that China is a leader in green patent activities; the number of patents increased from around 3000 in 2002 to over 20,000 by 2023. Meanwhile, the growth in India and Thailand was limited, while GRI activity in Brazil and Indonesia remained at relatively low levels. The data clearly show that ENT is closely linked to not only economic size but also to GEO, RQ, and GRI.

Building the theoretical basis

Inspired by the Porter Hypothesis (PH), this research recognizes RQ and GRI as channels through which economies respond to external geopolitical shocks. Developed by Michael Porter in the 1990s, the PH suggests that well-designed regulations can enhance innovation and long-term competitiveness (Porter, 1991). While traditional views see environmental policies as costly, the PH argues that proper regulations drive technological progress, resource efficiency, and improved environmental performance. The PH has three forms: weak, emphasizing the link between regulations and innovation (Jaffe & Palmer, 1997); strong, where innovations generate economic gains (Porter & van der Linde, 1995); and narrow, which depends on institutional conditions (Ambec, Cohen, Elgie, & Lanoie, 2013). More specifically, the PH provides the conceptual basis for understanding how RQ can transform environmental constraints into incentives for technological adaptation and structural energy transformation. Within this framework, GRI represents the technological response generated by effective regulatory systems, while the ENT reflects the broader structural outcome of this innovation process. Accordingly, the research adopts the strong version of the PH not merely as a general innovation argument, but as a dynamic institutional framework through which geopolitical conditions, RQ, and technological transformation interact during the ENT process.

However, ENT cannot be fully explained by RQ and GRI alone; geopolitical dynamics also play a critical role. Particularly in energy-import-dependent economies in the Global South, regional conflicts, supply disruptions, or sanctions can hinder the implementation of ENT policies. High GEO weakens investor confidence, drives up prices, and may lead governments to revert to fossil fuels for short-term security (Zhang & Usman, 2025). These conditions may weaken the RQ-innovation linkage predicted by the PH. On the other hand, in stable geopolitical contexts, regulatory predictability increases, which in turn boosts GRI and accelerates ENT. Hence, RQ promotes innovation through stability, GRI advances technology, and geopolitical stability influences both.

More importantly, the strong version of the PH provides a useful framework for understanding how geopolitical shocks may reshape the interaction between RQ and GRI within the ENT process. In the Global South context, geopolitical tensions do not operate solely as external disturbances affecting energy markets; they also influence institutional credibility, policy continuity, and investor expectations. Under elevated GEO, governments may redirect public resources toward short-term energy security priorities, weakening regulatory effectiveness and reducing incentives for long-term GRI investment. Consequently, geopolitical instability may indirectly hinder the ENT by disrupting the institutional conditions necessary for innovation-driven environmental transformation.

Within this framework, RQ can serve as a buffer against external geopolitical shocks. Economies with stronger RQ are generally better positioned to preserve policy consistency, maintain investor confidence, and sustain support for GRI during periods of geopolitical uncertainty. In contrast, weak institutional environments may amplify the adverse

effects of GEO by increasing policy unpredictability, discouraging private investment, and slowing technological adaptation. Thus, the link between GEO and ENT is not purely direct but also institutionally mediated. Accordingly, this paper conceptualizes the ENT process as an interconnected framework in which GEO influences both RQ and GRI dynamics, while RQ shapes the effectiveness of innovation-led transition policies. This interpretation extends the strong PH by incorporating geopolitical stability as a structural condition that influences the sustainability of innovation-driven ENT in the Global South.

Based on these relationships, Fig. 2 presents the study's conceptual framework, illustrating how GEO, RQ, and GRI interact to influence the ENT process in the Global South.

Research motivation, scope, and contributions

In the Global South, ENT represents a sensitive balance between EG, environmental sustainability, and geopolitical stability. Rapid urbanization and industrialization are driving up energy demand, while increasing dependence on fossil fuels is leading to higher carbon emissions and greater external energy vulnerability. Consequently, ENT has become a multidimensional policy challenge that requires balancing decarbonization objectives with energy security concerns amid increasing geopolitical uncertainty. Countries in the Global South remain particularly vulnerable to fluctuations in international energy prices, regional conflicts, and disruptions in global supply chains due to their continued dependence on imported fossil fuels and external finance. The Russia-Ukraine conflict further intensified these vulnerabilities by increasing energy market volatility and redirecting policy priorities toward short-term energy security concerns (Zhang & Usman, 2025).

At the same time, weak institutional and regulatory capacities continue to constrain REN deployment in many developing economies. Frequent policy changes, political instability, and limited institutional effectiveness reduce investor confidence and weaken long-term REN planning (Martinot, 1998). Further, limited domestic innovation capacity and continued dependence on foreign technology transfer slow the development of innovation-driven energy systems. In this regard, understanding the interconnected roles of RQ, GRI, and GEO becomes essential for evaluating the long-run sustainability of the ENT process in the Global South.

This article empirically analyzes the interactive effects of these factors on ENT across the five major Global South economies, covering the period from 2002 to 2023. Admittedly, the selected countries do not represent the entirety of the Global South. However, they constitute strategically important emerging economies characterized by high energy demand, rapid industrialization, increasing REN investments, and significant exposure to institutional and geopolitical challenges. Thus, the research seeks to provide context-specific insights into the structural dynamics of the ENT in major developing economies rather than universally generalizable conclusions for all Global South countries. The primary objective is to examine the dynamic relationships among GEO, RQ, GRI, and ENT. Based on this framework, the study aims to test the following hypotheses:

H1. Higher GEO is expected to constrain ENT by weakening policy credibility, increasing investment uncertainty, and shifting policy priorities toward short-term energy security.

H2. Higher RQ is expected to promote ENT by strengthening institutional credibility, policy consistency, and investment conditions for REN deployment.

H3. Higher GRI is expected to encourage ENT by enhancing technological capacity, energy efficiency, and the diffusion of REN.

Although the empirical model estimates direct long-run relationships, the hypotheses are interpreted within a broader institutional transmission framework. In this framework, RQ supports innovation-led

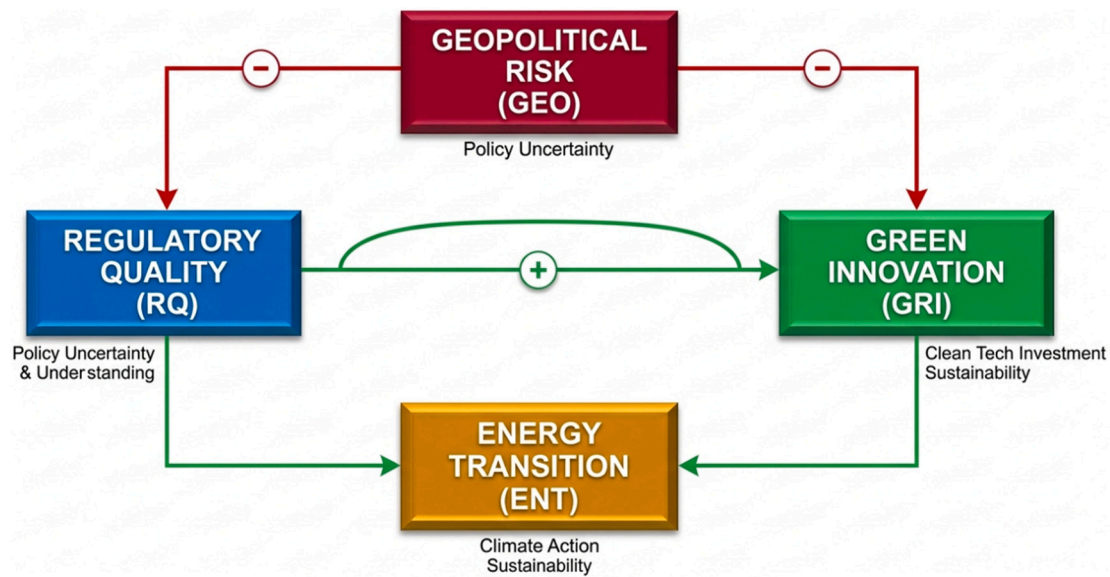


Fig. 2. Conceptual framework of the study
Source: Authors' own design.

ENT by strengthening policy credibility, regulatory consistency, and long-term investment incentives. In contrast, GEO may weaken the institutional conditions required for this mechanism to operate effectively. Accordingly, GEO may indirectly constrain ENT by undermining institutional effectiveness and reducing economies' capacity to sustain GRI. Conversely, stronger RQ may mitigate part of the adverse effects of geopolitical uncertainty by preserving policy credibility and supporting long-term innovation incentives.

This research contributes to the literature in three important ways. First, it develops an integrated sustainability governance framework to explain the REN transition in major Global South economies. Rather than treating GRI, RQ, and GEO as separate determinants, the study conceptualizes ENT as a process jointly shaped by technological capacity, institutional credibility, and external geopolitical uncertainty. This framing moves the analysis beyond conventional determinant-based explanations. Second, the study extends the strong version of the PH by identifying geopolitical stability as a structural condition for innovation-driven environmental transformation. This provides a broader political-economy reading of the Porter mechanism in developing economies. Third, the study provides context-specific evidence from five structurally important Global South economies characterized by rising energy demand, institutional constraints, fossil-fuel dependence, and exposure to external shocks. By focusing on this group, the paper highlights why ENT in developing economies cannot be understood solely as a technological process but also as a governance and resilience challenge amid geopolitical uncertainty.

The remainder of the paper is structured as follows: Section "Literature review" reviews the relevant literature; Section "Data and method" discusses the data and method; Section "Results and discussion" presents the empirical results; and Section "Conclusion" outlines the policy implications, particularly regarding the achievement of SDG 7 in the Global South.

Literature review

Numerous factors influence the pace of ENT, a fundamental prerequisite for ecological sustainability. ENT success is closely related to key elements such as a country's level of development, globalization, geopolitical relations, technological progress, and institutional quality. The following literature review summarizes findings from studies that specifically analyze the links among ENT, GEO, GRI, and RQ.

The influence of geopolitical risk on the energy transition

Previous studies suggest that GEO highlights energy security vulnerabilities and can thus play a key role in accelerating the ENT process by encouraging investment in domestic clean energy sources. For example, Hille's (2023) study of 37 EU countries (1991–2021) showed that increases in GEO in supplier countries led to greater adoption of REN in the EU. Consistently, Wang, Zhang, Zhang, and Feng (2025), using results from 39 countries (2003–2019), revealed that GEO encouraged the transition to REN, with the effect more pronounced in environments with more diverse financial incentives, higher energy efficiency, and more effective government management. Furthermore, He et al. (2025) showed that increases in GEO across 41 nations (2000–2021) led to faster growth in ENT. This relationship is particularly robust in countries with low levels of TR, oil revenues, and technological innovation.

On the other hand, many studies suggest that GEO may hinder progress by discouraging long-term investments, disrupting supply chains, and shifting policy focus away from long-term climate goals. In this regard, analyzing 119 economies (2014–2022), Shittu, Saqib, Tang, and Chen (2025) reported that ENT is negatively associated with GEO across all quantiles, and this correlation is particularly strong in less developed countries that are heavily reliant on fossil fuels and lack energy diversification and financial resources. Similarly, Lee and Lee (2024) analyzed 30 Chinese provinces (2000–2019) and showed that GEO negatively impacts REN development, especially in less-developed western regions, where weaker economic and social conditions increase these effects. Meanwhile, Zhu, Hunjra, Alharbi, and Zhao (2025) examined 41 countries (2003–2025) and asserted that GEO delays ENT by disrupting supply chains, rising price volatility, and shifting policy priorities. This effect is less pronounced in economies with strong financial systems, adaptable labor forces, and advanced REN, but more significant in militarized and resource-dependent countries.

In sum, the relationship between GEO and ENT remains ambiguous. Some studies argue that GEO may accelerate REN investments by increasing energy security concerns. Others emphasize the disruptive effects of uncertainty, supply-chain instability, and investment volatility. These conflicting findings suggest that GEO's impact depends heavily on institutional resilience, technological preparedness, and structural energy dependence, particularly in developing economies.

The role of green innovation in energy transition

The GRI, covering eco-innovation, green technologies, and green patents, is widely recognized as a key driver of global ENT. Several studies provide substantial evidence that the GRI has a direct and positive influence on ENT, resulting in increased REN capacity, enhanced efficiency, and reduced costs. Consistently, [Wei, Wen, Wang, et al. \(2023\)](#), analyzing 49 APAC economies (2000–2019), found that GRI promotes REN, reduces energy intensity, and improves access to clean cooking. It supports industrial modernization and expands renewable capacity. Similarly, [Tiwarei, Shahzad, Alofaysan, Walsh, and Kumari \(2024\)](#), in the United States (2008–2022), demonstrated that GRI has a positive influence on ENT in both the short and long term. [Shahzad's \(2025\)](#) analysis of 49 high-income countries (1990–2023) also confirmed that GRI leads to a significant increase in clean energy use and a decrease in dirty energy use. Moreover, in their analysis of OECD economies (1990–2017), [Li, Zhang, Ali, and Khan \(2020\)](#) identified GRI as a significant factor positively influencing ENT, alongside energy productivity, human capital, income, and energy prices. Similarly, [Zhang, Alharthi, Ali, Abbas, and Taghizadeh-Hesary \(2022\)](#), analyzing OECD countries (1991–2019), concluded that GRI, human capital, and energy prices are the primary drivers of rising REN consumption and declining fossil fuel use.

Furthermore, research shows that financial mechanisms and environmental policies often influence the effectiveness of GRI. In this context, [Lee, Wang, and Chang \(2023\)](#) found that in China (2001–2019), green finance directly supported REN and indirectly supported it by boosting research and development (R&D). [Muhammad and Hoffmann \(2024\)](#) analyzed 16 German states (2008–2021) and found that R&D funding alone did not support ENT, whereas R&D combined with patent applications significantly boosted ENT, highlighting the importance of financial investment for technological progress. [Maghyereh, Boulanouar, and Essid \(2025\)](#) found a stable, positive link between environmental policy stringency and ENT across 16 countries (2004–2019); however, the effect of GRI on ENT varies over time and declines after 2014 as GRI matures. [Le, Mai, Nguyen, Nguyen, and Do \(2025\)](#) analyzed OECD countries (1994–2020) and found that environmental taxes positively affect ENT in the lower and middle quantiles, while financial inclusion positively affects ENT across all quantiles. GRI tends to have a positive influence on ENT, though not significantly, across all quantiles.

The role of regulatory quality on energy transition

The quality of a nation's institutions and regulatory framework is a crucial factor in its ability to navigate ENT successfully. RQ, the ability to develop and enforce effective policies, uphold the rule of law, control corruption, and enhance bureaucratic efficiency, creates the necessary environment for REN investments and the phasing out of fossil fuels. The available empirical evidence strongly indicates that RQ directly benefits ENT by fostering a stable, predictable, and supportive environment. For example, [Uzar's \(2020\)](#) study of 38 countries (1990–2015) found that RQ positively influences long-term REN consumption, suggesting that effective institutions prevent corruption and instability, creating a stable policy environment for ENT. Likewise, [Akorli and Adom's \(2023\)](#) study of 46 African countries revealed a direct link between RQ and energy efficiency: improving RQ and reducing corruption can help African countries transition out of low energy efficiency, a key aspect of ENT. In parallel, [Olaniyi, Al-Faryan, and Ogbaro \(2025\)](#), for African countries (1990–2019), reported that weak institutions enable rent-seeking in finance, diverting resources to dirty energy and undermining ENT. The study found that most African countries fall below RQ threshold, indicating that their financial development has not yet been effectively directed toward clean energy. Moreover, [Pata, Alola, Erdogan, and Kartal \(2023\)](#), in their study of G7 countries (2004–2018), found that government efficiency and RQ are not significantly associated with ENT,

whereas EG shows a positive effect. Conversely, economic policy uncertainty, urbanization, and GEO negatively impact ENT, suggesting that in highly developed countries with a high baseline RQ, these factors may be the primary constraints. In contrast, RQ may be a hindering factor in developing countries. For example, in a study of five Asian economies (China, India, Japan, Singapore, and Malaysia) (1995–2021), [Xu, Chen, and Dong \(2024\)](#) indicated that globalization, financial technologies, and oil revenues have a positive effect on ENT, while RQ harms it. This suggests that in certain contexts, particularly in rapidly developing economies, specific policy priorities can initially support energy security and EG rather than a rapid ENT, highlighting the importance of RQ.

While the existing literature offers significant insights into the individual impacts of GEO, GRI, and RQ on ENT, several key gaps remain unresolved. First, previous studies have generally examined these determinants in isolation, rarely conceptualizing ENT as a result of the simultaneous interaction of technological, institutional, and geopolitical dynamics. Second, while the literature often focuses on advanced economies or large heterogeneous samples, structurally heterogeneous Global South economies facing energy security pressures, institutional fragility, and technological dependence have received relatively limited attention. Third, despite the increasing importance of geopolitical uncertainty, the extent to which GEO can indirectly influence ENT through institutional and innovation channels has not been sufficiently emphasized. In contrast, this research develops an integrated analytical framework based on a robust version of the PH by examining GEO, RQ, and GRI together in major Global South economies. This framework reveals how the simultaneous interactions of institutional, technological, and geopolitical conditions shape ENT.

Data and method

Sources of data

The article analyzes the association between GEO, RQ, and GRI and ENT in the five major Global South nations, namely Brazil, China, India, Indonesia, and Thailand, over the 2002–2023 period, controlling for EG and trade openness (TR). However, the direction of this link may depend on the technological content and energy intensity of trade structures, particularly in developing economies where industrial production is reliant on fossil fuels. The 2002–2023 analysis period was chosen because WGI first published RQ data in 2002, and it remained available through 2023. This timeframe also coincides with a period of significant change in global energy markets, symbolized by increasing environmental regulations and the widespread adoption of green technologies.

The five countries selected for this research were chosen since they represent structurally significant developing economies in the Global South with substantial REN potential. These economies have distinct industrialization processes, institutional structures, energy dependency patterns, and technological capacities, enabling the analysis to capture the heterogeneous transition dynamics across large developing economies. Furthermore, as they collectively represent a significant share of the Global South's population, industrial output, energy consumption, and carbon emissions, they are particularly important for examining the structural determinants of the ENT process. However, the findings should be interpreted as evidence from a strategically selected group of major emerging economies, not as outputs fully representative of the entire Global South.

ENT measures the share of REN in total primary energy, from [OWD \(2025\)](#). GRI shows the number of environment-related patents from the [OECD \(2025\)](#). RQ indicator, ranging from -2.5 to $+2.5$, is derived from the [WGI \(2025\)](#). EG is real GDP per capita (constant 2015 US dollars), and TR is the total trade-to-GDP ratio (%), both sourced from the World Development Indicators ([WDI, 2025](#)). GEO data are taken from [Caldara and Iacoviello \(2022\)](#)'s Geopolitical Risk Index. This comprehensive dataset provides a robust empirical basis for demonstrating how

institutional, technological, and geopolitical dynamics relate to the ENT pathways of leading Global South economies.

Theoretical background of the variables and model description

In the literature, various measures have been used to summarize the multifaceted nature of the ENT process, including the share of REN in total consumption (Lopez, Sanagustín-Fons, & Fierro, 2026), energy intensity (Li, Lin, & Su, 2025), carbon emission intensity (Bai et al., 2023), the share of fossil fuels, and composite indices (McCauley & Pettigrew, 2023). However, each represents only a partial dimension of the energy system. For example, energy or carbon intensity measures efficiency and emissions, but cannot measure structural change. Fossil fuel shares and composite indices often suffer from data inconsistencies, especially in developing economies. Hence, this study uses the share of REN in total primary energy use as the leading indicator of ENT, as it effectively captures structural changes and the degree of decarbonization in national energy portfolios. Although ENT is multidimensional, the share of REN in total primary energy consumption is widely used as a standard indicator due to its cross-country comparability and data availability. This variable captures structural changes in the energy mix and provides a consistent indicator of progress in decarbonization.

The key explanatory variables (GEO, RQ, and GRI) represent the political, institutional, and technological aspects of ENT. GEO is a media-based indicator of energy security and investment volatility (Zhang & Usman, 2025). RQ demonstrates institutional capacity for policymaking that enhances market efficiency and encourages private-sector participation. GRI, measured as environmental patents, shows the tangible outcomes of environmental innovation (Çamkaya, Kaya, & Karabayir, 2025). The article is theoretically based on PH (Porter & van der Linde, 1995), arguing that effective regulations enhance competitiveness by promoting innovation (Caglar, Demirdag, Syed, & Destek, 2026; Caglar, Ulug, & Demirdag, 2026). In this model, RQ serves as the institutional pathway to policy effectiveness, GRI as the technological outcome, and GEO as the external factor influencing energy security, investment flows, and technology transfer. So, high RQ and GRI can accelerate ENT even in the presence of significant GEO.

To account for macroeconomic effects, EG and TR are included as control variables. While increased income may initially lead to increased fossil fuel consumption, sustainable growth incentivizes investment in REN (Özbek & Naimoğlu, 2025). TR facilitates the transition by easing the diffusion of technology and capital flows, but remains vulnerable to global price shocks (Józwik, Sarıgül, Topcu, Çetin, & Doğan, 2025).

Based on this framework, the research analyzes the association of GEO, RQ, and GRI with ENT, controlling for EG and TR, using the following panel regression model:

$$ENT_{it} = \alpha_i + \beta_1 GEO_{it} + \beta_2 RQ_{it} + \beta_3 GRI_{it} + \beta_4 EG_{it} + \beta_5 TR_{it} + \varepsilon_{it} \quad (1)$$

where ENT_{it} shows the level of ENT in country i at time t ; α_i denotes country-specific, time-invariant effects such as institutional structure or geography; and ε_{it} is the error term. While GRI and EG are converted to logarithmic form, ENT, RQ, GEO, and TR remain at their original scales. ENT is treated as a ratio variable instead of a logarithmic conversion, as it is measured as the share of REN in total primary energy. RQ and GEO are index-based variables, and TR is measured as a ratio. Accordingly, Eq. (1) is examined based on the theoretical framework discussed above. The main reasons for excluding additional variables are data availability constraints and the need to maintain a balanced panel structure across countries and over time. In particular, comparable long-term data for these variables are not available for all selected Global South economies during the study period. Therefore, the model prioritizes key structural determinants that are consistently measurable and comparable across countries.

Econometric tools

The research employed advanced panel data techniques to ensure accurate estimation and address cross-sectional dependence (CD) and slope heterogeneity (SH). First, diagnostic tests for interdependence and heterogeneity were conducted, followed by the running of estimators incorporating cross-country structural dynamics and common shocks. The paper employed the Pesaran CD test (Pesaran, 2004) and Breusch-Pagan LM test (Breusch & Pagan, 1980) to determine whether shocks in one country affect other countries. Due to similar energy policies and growth patterns among Global South economies, the probability of CD is high, and neglecting this could distort the results. To test SH, the Pesaran and Yamagata (2008) test, which detects heterogeneity among slope coefficients, was performed, justifying the use of second-generation estimators that account for this heterogeneity and better capture the structural dynamics of the ENT. Next, to assess stationarity, the Pesaran (2007) Cross-Sectionally Augmented IPS (CIPS) unit root test was run, considering CD. The CIPS test equation is given as follows:

$$\Delta y_{it} = \alpha_i + \beta_i y_{i,t-1} + \gamma_i \bar{y} + \delta_i \Delta \bar{y} + \varepsilon_{it} \quad (2)$$

where, y_{it} represents ENT series for country i , $\bar{y}$ is the cross-sectional mean, Δ denotes the difference operator, α_i is the constant term, and ε_{it} is the error term. A statistically significant and negative β_i coefficient indicates stationarity.

The CIPS test provides results that better reflect panel dynamics by accounting for inter-country heterogeneity and common shocks. After confirming that all variables have the same degree of integration, the Durbin-Hausman panel cointegration test was used to show the presence of a long-term equilibrium link. Unlike alternatives such as the Westerlund (2008) test, the Durbin-Hausman method is more effective in small samples and maintains its reliability even with CD. The test is defined as follows:

$$y_{it} = \alpha_i + \beta_i' x_{it} + e_{it} \quad (3)$$

where y_{it} denotes ENT, x_{it} is the vector of explanatory variables (GRI, RQ, GEO, EG, TR), and e_{it} represents the residuals.

A stationary error term indicates cointegration and minimizes bias and endogeneity issues compared to Engle-Granger or Pedroni tests. For long-term estimation, the Common Correlated Effects Fully Modified (CUP-FM) estimator by Bai, Kao, and Ng (2009) was used because it accounts for CD and unobserved common factors:

$$y_{it} = \alpha_i + \lambda_i' \beta_i + \lambda_i' F_t + \varepsilon_{it} \quad (4)$$

Here, F_t reflects unobserved common factors, λ_i shows the country-specific loadings, and β_i denotes the long-run coefficient vector. This approach reduces estimation bias by assuming that countries are exposed to similar macroeconomic shocks affecting ENT. To increase robustness, the Bias-Adjusted (CUP-BC) estimator proposed by Bai et al. (2009) was also used:

$$\hat{\beta}_{BC} = \hat{\beta}_{FM} - \hat{B}/T \quad (5)$$

where $\hat{\beta}_{FM}$ is the CUP-FM estimate, $\hat{B}$ is the bias correction term, and T reflects the time dimension. This correction improves estimation efficiency in smaller panels and provides more reliable long-term coefficients for ENT. As an additional robustness check, the Cross-Sectionally Augmented Autoregressive Distributed Lag (CS-ARDL) estimator proposed by Chudik and Pesaran (2015) was employed, allowing for both short- and long-term dynamics under CD and SH. Finally, directional predictive relationships between variables were investigated using the Dumitrescu and Hurlin (2012) panel Granger non-causality test for heterogeneous panel settings:

$$y_{it} = \alpha_i + \sum_{k=1}^K \gamma_{ik} y_{i,t-k} + \sum_{k=1}^K \beta_{ik} x_{i,t-k} + \varepsilon_{it} \quad (6)$$

In this model, y_{it} denotes the dependent variable (ENT), x_{it} is the explanatory variable (e.g., GRI, RQ, GEO), and γ_{ik} , β_{ik} the lagged effects. The flexibility of the test allows for heterogeneous predictive relationships across countries, offering a more realistic framework than traditional Granger tests. In summary, the paper integrated second-generation panel data techniques, including CIPS, Durbin-Hausman, CUP-FM/BC, and Dumitrescu-Hurlin methods, to account for heterogeneity, CD, and long-term dynamics among Global South economies. This approach ensures both statistical robustness and policy appropriateness in the analysis of long-run correlates of ENT.

The reason for using second-generation panel estimators is the presence of CD, SH, and long-term interdependence among the selected economies. Additionally, potential endogeneity may arise from reverse causality, omitted variable bias, and unobserved common shocks. For example, GRI and RQ may support renewable ENT, but progress in REN deployment may also encourage further innovation and institutional reforms. Similarly, GEO and TR may be shaped by broader macroeconomic and policy conditions. To mitigate these concerns, the study employed second-generation panel estimators. CUP-FM and CUP-BC are particularly useful in cointegrated panels as they correct for endogenous variability and serial correlation, while CS-ARDL further captures dynamic adjustment. Therefore, the results reveal robust long-term relationships rather than definitive causal effects.

Results and discussion

Summary of the preliminary analysis results

Before estimating long-term relationships, several pre-tests were conducted to validate the strength of the empirical analyses. The findings shown in Panel A in Table 1 explicitly reject the null hypothesis of CD at the 1% significance level for all variables, implying that the selected Global South economies are interconnected, likely due to shared global economic shocks, regional policy spillovers, and integrated energy markets, meaning that economic developments in one country may create spillover effects in others, making second-generation panel estimators suitable for empirical analysis. Additionally, the results in Panel B reject the null hypothesis of SH at the $\alpha = 0.01$ level, revealing that the effects of the explanatory variables on ENT differ between countries due to heterogeneity in economic structures, RQ, innovation capacity, and geopolitical exposure.

Subsequently, the Pesaran (2007) CIPS results in Table 2, Panel A, showed that all variables were not stationary at their levels, but became stationary after applying first differences and confirmed their first-order integration (I(1)). As seen in Table 2, Panel B, the Durbin-Hausman

Table 1
Dependence and slope homogeneity findings.

Variable	Breusch & Pagan, 1980		Pesaran, 2004	
	LM-stat	p-value	CD-stat	p-value
Panel A: Cross-sectional dependence findings				
ENT	149.803*	0.000	31.261*	0.000
GRI	84.797*	0.000	16.725*	0.000
RQ	40.008*	0.000	6.710*	0.000
EG	188.911*	0.000	40.005*	0.000
GEO	42.262*	0.000	7.214*	0.000
TR	49.277*	0.000	8.782*	0.000
Panel B: Slope homogeneity findings				
tests	stat.	p-value		
Delta	7.011*	0.000		
adj.	8.491*	0.000		

Note: * indicate Significant value at 1%.

Table 2
Unit root test and cointegration results.

Variables	CIPS	
	Level	First differences
Panel A: Pesaran (2007) findings		
ENT	−1.610	−5.070
GRI	−1.997	−4.165
RQ	−2.009	−4.386
EG	−1.595	−2.954
GEO	−3.186	−5.478
TR	−1.629	−3.183
Panel B: Cointegration tests finding		
	Value	p-value
lm statistic	9.898 ^a	0.794
Durbin-g	−1.688 ^a	0.046
Durbin-p	−1.316 ^a	0.094

Note: Critical values for CIPS test statistics: 10% -2.21, 5% -2.33, and 1% -2.57. a indicates cointegration.

cointegration test results indicated the existence of a consistent long-term relationship between ENT and all other variables.

The CUP-FM and CUP-BC outputs in Table 3 showed that GRI, RQ, and EG were significantly and positively associated with ENT. In contrast, GEO and TR were negatively associated with ENT in the long term. The CUP-FM and CUP-BC outputs are consistent with the proposed hypotheses. The findings were robust across the CUP-FM and CUP-BC estimators, as both approaches produced consistent coefficient signs and significance levels for all variables.

The consistency between the CUP-FM and CUP-BC results strengthened the robustness of the estimated relationships. It confirmed that the direction and magnitude of the effects are not sensitive to the choice of estimator. In particular, the larger coefficient of RQ indicated that institutional quality had a significant long-term correlation with ENT. The Dumitrescu and Hurlin (2012) panel test was used to examine the directional predictive relationships between explanatory variables and ENT. The results in Table 4 showed predictive links from GRI, RQ, EG, GEO, and TR to ENT. These findings were consistent with the CUP-FM and CUP-BC results and showed that the variables were long-term correlated with ENT in large Global South economies. Therefore, the results provided additional empirical support for policy discussions focusing on innovation, RQ, GEO management, economic capacity, and trade structure in the ENT process.

The CS-ARDL estimates, reported in Table 5, further corroborate the CUP-FM and CUP-BC findings, thereby confirming the robustness of the baseline results. In the long term, GRI, RQ, and EG were positively correlated with ENT, whereas GEO and TR were negatively correlated with ENT. The consistency of coefficient signs among alternative

Table 3
CUP-FM and CUP-BC results.

Variables	Coefficient	p-value
Panel A: CUP-FM		
GRI	0.305*	0.000
RQ	2.433*	0.000
EG	4.867*	0.000
GEO	−0.474*	0.000
TR	−0.096*	0.000
Panel B: CUP-BC		
Variables	Coefficient	p-value
GRI	0.321*	0.000
RQ	1.154*	0.000
EG	3.740*	0.000
GEO	−0.906*	0.000
TR	−0.067*	0.000

Note: * indicate Significant value at 1%.

Table 4
Dumitrescu and Hurlin (2012) panel Granger non-causality test.

Null hypothesis	W-stat	Zbar-stat	p-value	Directional link
GRI does not predict ENT	7.615	1.848	0.064	+
RQ does not predict ENT	4.192	5.048	0.000	+
EG does not predict ENT	14.635	6.813	0.000	+
GEO does not predict ENT	10.349	3.782	0.000	+
TR does not predict ENT	3.201	3.347	0.000	+

Note: The null hypothesis states that the explanatory variable does not Granger-predict ENT.

estimators indicated that the underlying findings were not sensitive to estimator choice and remained robust even after accounting for CD and dynamic adjustment. The significant and negative error correction term (ECT) confirmed the existence of a stable long-term equilibrium relationship between the variables. Statistically insignificant short-term coefficients suggested that the effects of innovation, institutions, GEO, and TR emerged primarily through long-term structural channels rather than immediate short-term responses.

Discussion of main findings

Empirical findings demonstrate that the ENT process in the Global South is shaped not only by technological advancement but also by institutional quality and geopolitical conditions. In this context, the results support the broader rationale of PH, suggesting that effective regulatory structures can foster green innovation (GRI) and facilitate long-term structural transformation (ENT). At the same time, the findings reveal that innovation-driven ENT effectiveness remains highly sensitive to geopolitical instability and institutional resilience. Accordingly, ENT in emerging economies should be understood as a multidimensional socio-economic process shaped by the interplay among technological capability, institutional quality, and geopolitical pressures. In conclusion, the findings confirm all hypotheses and reveal that GEO weakens ENT, while RQ and GRI support it.

The constructive role of green innovation and regulatory quality

In the context of strong PH, the positive coefficients of GRI and RQ demonstrate that environmental and institutional policies support long-term structural transformation through innovation-boosting mechanisms. Empirical findings support *H2* and *H3*, as well as the argument that well-functioning regulatory systems not only impose environmental costs but can also encourage technological adaptation and enhance the sustainability of energy systems. The evidence clearly shows that GRI significantly drives ENT in the Global South, confirming *H3*. The magnitude of the estimated coefficients further emphasizes the relative importance of these factors. In particular, the strong positive coefficient on RQ among the estimators, supporting *H2*, indicates that institutional credibility has a significant, long-term relationship with ENT. This finding supports the argument that conversion to REN use is more likely when stable regulatory frameworks and reliable policy environments support GRI. This finding corroborates Wei et al. (2023)'s finding that GRI directly advances ENT in APEC economies and Tiwari et al. (2024)'s

confirmation of its positive association in the US. Specifically, GRI is a critical tool for ENT as it directly facilitates the decarbonization of the energy system through new technologies, reduces reliance on fossil fuels, and increases energy efficiency (Shahzad, 2025). Furthermore, some studies show that eco-innovation reduces reliance on non-REN sources and increases the use of REN (Li et al., 2020), while technological advancements improve the efficiency of alternative energy sources, thus enhancing green ENT (Maghyereh et al., 2025). While this positive association is generally consistent with the existing literature, the current findings offer a more nuanced interpretation, suggesting that GRI's contribution becomes structurally dependent on RQ and geopolitical conditions. In other words, technological innovation alone may not automatically yield a successful ENT unless it is supported by a stable regulatory environment and protected from geopolitical disruptions. This interconnected interpretation represents a significant departure from traditional approaches that often assess the impacts of innovation in isolation.

However, transition dynamics are not entirely homogeneous across the selected economies. For example, China exhibits a significantly stronger innovation-driven transition model due to its higher intensity of green patent activity, stronger industrial capacity, and more extensive state-backed REN investments. In contrast, economies such as Indonesia and Thailand exhibit relatively slower transition dynamics, reflecting differences in institutional efficiency, technological readiness, and external vulnerability. These heterogeneous characteristics underscore the importance of country-sensitive policy approaches within the broader Global South context.

Furthermore, the RQ coefficient is highly positive, indicating that robust regulatory frameworks are vital to translating GRI into real-world transition outcomes. RQ plays a key role here, as it helps create a stable and predictable investment environment necessary to attract long-term capital and mitigate risks for large-scale REN projects (Olaniyi et al., 2025). This finding supports Uzar's (2020) findings and is also backed by broader research (e.g., Wang, Wang, and Wang (2023) found a positive link between RQ and ENT in Asian economies). In the Global South, where institutional frameworks are generally at low to medium levels, improving RQ is highly relevant. As Akorli and Adom (2023) suggest, removing bureaucratic hurdles and reducing corruption, two key obstacles to energy efficiency, may boost investor confidence, thus enabling the effective adoption and use of green technologies. The synergy between GRI and RQ, supporting strong PH, suggests that well-designed regulations can spur innovation and deliver economic and environmental benefits (Herman & Xiang, 2022). In the Global South, RQ serves as a crucial conduit for converting regulatory signals to GRI and accelerating ENT. This institutional role is particularly important in the Global South, where market imperfections, funding constraints, and policy discontinuities frequently weaken the effectiveness of innovation systems. Unlike developed economies with mature institutional structures, developing economies require stronger regulatory coordination to translate technological potential into real ENT outcomes. Therefore, the positive contribution of RQ in this study reflects not only improved governance performance but also the broader institutional capacity necessary to sustain innovation-driven structural transformation.

Table 5
Robustness check using the CS-ARDL estimator.

Long-run				Short-run			
Variables	Coeff.	Std. err.	P-value	Variables	Coeff.	Std. err.	P-value
GRI	0.183*	0.033	0.000	ΔGRI	0.129	0.233	0.579
RQ	1.613*	0.286	0.000	ΔRQ	1.139	8.178	0.889
EG	2.366*	0.420	0.000	ΔEG	1.670	35.545	0.963
GEO	-2.137*	0.379	0.000	ΔGEO	-1.509	7.023	0.830
TR	-0.034*	0.006	0.000	ΔTR	-0.024	0.104	0.815
				ECT	-0.799*	0.133	0.000

Note: ECT denotes the error-correction term. *Significant value at 1%.

The disincentive effect of geopolitical risk

A key result of this study is a long-run negative nexus between GEO and ENT, which strongly supports the *H1*. The results show that rising geopolitical tensions hinder the adoption of REN in these emerging economies. This is consistent with existing research, including Lee and Lee (2024) and Shittu et al. (2025), which suggests that GEO is negatively associated with ENT across all percentiles, especially in less-developed countries or regions. The Russia-Ukraine War serves as a clear example, causing a sharp increase in global GEO, major fluctuations in energy supplies, exposure of supply chain vulnerabilities, and record-high energy prices (Wang et al., 2023). Additionally, ongoing tensions among major powers, such as the trade and technology dispute between the US and China, have exacerbated GEO, causing supply chain disruptions for critical materials like rare earths and creating an atmosphere of uncertainty that is hampering investments in clean energy (Depraite & Goutte, 2025). This conflict has directly slowed the adoption of REN, as funds allocated for REN projects have been used to support struggling businesses and meet military needs (Zhao et al., 2024).

The reasons for this outcome are complex. First, GEO creates a highly unpredictable environment, discouraging long-term, capital-intensive investments needed for REN projects. Pata et al. (2023) suggest that any rise in GEO makes the success, payback, and profitability of investments uncertain, prompting investors to choose more stable options and delaying clean energy investments. Second, during periods of rising GEO, governments might prioritize short-term energy security over long-term sustainability goals. This could lead to continued reliance on fossil fuels to meet immediate supply needs, a pattern that may persist for some time. This was evident in Europe after the Russia-Ukraine War, where some countries, including Germany, with ambitious ENT targets, continued to rely on coal-fired power generation (Wang et al., 2023). These conditions serve as a warning for energy-importing countries like India and Indonesia, which are highly vulnerable to fluctuations in global energy prices and supply chain disruptions. Overall, data from the Global South indicate that investor uncertainty, supply chain issues, and GEO-related policy changes have delayed the shift to ENT.

From a theoretical perspective, these findings also support the institutional rationale underlying the robust PH. The PH assumes that firms increase innovation and productivity by responding positively to clear, reliable, and stable regulatory signals. However, amid increasing geopolitical uncertainty, the stability of policy expectations can be significantly disrupted. In the Global South, where institutional resilience is generally more limited, geopolitical shocks can weaken regulatory credibility, increase investment uncertainty, and reduce the long-term incentives necessary for GRI. Consequently, the innovation-boosting mechanism proposed by the PH may become less effective in politically unstable environments, thus slowing down the innovation process.

Accordingly, the findings suggest that GEO should not be interpreted solely as an external macroeconomic disruption affecting energy markets. Rather, GEO may also weaken the institutional credibility necessary to sustain long-term GRI and RQ. This interpretation offers an additional theoretical perspective by highlighting the indirect institutional transmission channels through which geopolitical uncertainty can affect the ENT process in developing economies. This result also demonstrates that the explanatory power of the PH may be context-dependent. While the theory assumes that firms respond favorably to stable, reliable regulatory incentives, the findings show that such innovation-boosting mechanisms may weaken under persistent geopolitical instability and institutional fragility. Hence, the study highlights an important limiting condition of the PH in the context of developing economies exposed to external geopolitical shocks.

The dual role of economic growth and trade openness

In the Global South, the connection between EG and ENT is both positive and significant. This shows that economic prosperity provides

the necessary fiscal space, technological ability, and investment funds to support REN infrastructure for ENT. These findings align with studies on G7 and OECD countries, where EG has been a primary driver of ENT (Pata et al., 2023; Zhang et al., 2022). Clearly, stronger economies are better able to subsidize REN, invest in R&D, and cover initial costs for ENT. Conversely, the relationship between TR and the ENT appears more complex and structurally conditional in the context of the Global South. Although TR is generally expected to facilitate the diffusion of clean technologies, capital flows, and environmental knowledge, the empirical findings indicate a significant negative association with ENT. One possible explanation is that the trade structures of many developing economies remain heavily concentrated in energy-intensive manufacturing, fossil fuel-dependent industrial production, and resource-based exports. In such contexts, increasing TR may expand industrial output and transportation activities that rely predominantly on conventional energy sources, reinforcing carbon-intensive production patterns.

Conclusion

Theoretical/analytical implications

Strict and well-designed environmental regulations may enhance environmental performance and long-term economic competitiveness by incentivizing firms to innovate. This idea forms the basis of the PH. Initially, firms face time and cost barriers due to technological changes, but they then innovate to increase efficiency. As a result, they maximize their profits by gaining a competitive advantage. However, geopolitical upheavals such as war, energy supply disruptions, and trade embargoes can increase concerns about energy security. Countries respond by adopting ENT to reduce reliance on external sources, secure energy supply, and reduce economic vulnerability. Consequently, the innovation and competitive gains suggested by the PH are driven by micro-level environmental policies and macro-level geopolitical forces.

The findings expand on the concept of strong PH by highlighting that the effectiveness of innovation-driven environmental transformation depends not only on environmental regulations but also on the institutional capacity to sustain them amid geopolitical uncertainty. In this sense, geopolitical stability and regulatory resilience jointly determine whether GRI can effectively accelerate the ENT process. Accordingly, the study contributes to the theoretical literature by demonstrating that the PH should be interpreted within a broader political economy framework in developing economies. The effectiveness of regulation-driven innovation depends not only on policy design but also on the institutional and geopolitical environment in which these policies operate.

This research attempts to understand how RQ, GRI, TR, EG, and GEO relate to ENT in a panel of five major economies in the Global South. Empirical results, obtained using rigorous methods and robustness checks such as CUP-FM/BC, Dumitrescu-Hurlin, and CS-ARDL estimators, show that GRI, RQ, and EG are positively associated with ENT. At the same time, GEO and TR are negatively correlated in the long term. The findings offer analytical guidance for policy design aligned with SDG-7 goals. However, achieving these goals depends on broader structural, institutional, and geopolitical conditions that are outside the scope of this study. More importantly, the results suggest that the dynamics of ENT in the major emerging economies of the Global South are structurally different from those in developed economies. Institutional resilience, geopolitical stability, and technological absorption capacity together determine whether innovation-driven environmental transformation can succeed under external shocks and development constraints. Accordingly, the key contribution of this study is not only to identify the determinants of ENT but also to demonstrate that the effectiveness of global innovation is critically dependent on the broader institutional and geopolitical environment. This integrated perspective contributes to the political economy literature on sustainable ENT in

developing economies. Overall, the study provides an integrated empirical framework that simultaneously examines GEO, RQ, and GRI within a unified long-run panel structure for major Global South economies, offering a more comprehensive understanding of sustainable ENT dynamics under external uncertainty.

Policy implications

The findings offer several policy implications for the major Global South economies examined in this study. Since the estimates were obtained at the panel level, these implications should be read as common long-term policy directions rather than country-specific prescriptions.

First, the negative relationship between GEO and ENT suggests that REN policies should be designed with greater attention to risk management. External shocks can delay REN investments by increasing uncertainty, weakening investor confidence, and disrupting supply chains. Thus, governments may enhance the resilience of REN projects through long-term policy commitments, political risk guarantees, green financing tools, and stronger cooperation with development banks.

Second, the positive role of RQ reveals the importance of institutions for ENT. Investors may be reluctant to invest in REN if they face unstable regulations, unclear licensing procedures, or weak policy implementation. Therefore, major emerging economies may strengthen regulatory transparency, provide predictable investment rules, and maintain credible long-term REN roadmaps. Stronger institutions can help transform GRI into genuine REN implementation.

Third, the positive association between GRI and ENT underscores the need to support local technological capacity. Public R&D funding, university-industry collaboration, intellectual property protection, and green technology transfer can be strengthened. In economies with relatively high innovation capacity, such as China, policy may focus on commercializing green technologies and improving grid integration. In economies with lower innovation capacity, such as Indonesia and Thailand, attracting green foreign investment and establishing local innovation systems can be a priority.

Fourth, the negative relationship between TR and ENT demonstrates that trade liberalization alone does not guarantee cleaner energy systems. If trade remains linked to fossil fuel-intensive production or carbon-intensive imports, it can slow the development of ENT. Therefore, trade policy should be aligned with sustainability goals. This may include revising tariff structures that support fossil fuel imports, supporting the import of REN technologies, and adding green technology clauses to regional and bilateral trade agreements.

Finally, the positive role of EG should not be interpreted as a call for growth at any cost. Rather, economic expansion can support ENT when it is directed toward low-carbon infrastructure, REN investment, grid modernization, and green skills. For economies with relatively high REN shares, such as Brazil, diversification beyond hydropower is crucial. For rapidly growing and fossil fuel-dependent economies like India and Indonesia, REN tenders, green bonds, and public-private partnerships can help mobilize investment.

Overall, the results suggest that progress toward SDG-7 requires more than technological progress. It depends on reliable institutions, innovation capacity, green-focused trade policies, and the ability to protect REN investment from geopolitical uncertainty.

Limitations and future research directions

The following limitations should be considered when evaluating the outcomes: First, although the selected countries are structurally important major Global South economies, the sample does not represent the full diversity of the Global South. Brazil, China, India, Indonesia, and Thailand differ significantly in terms of REN capacity, institutional quality, innovation systems, industrial structure, and geopolitical exposure. Hence, the results should be interpreted as evidence from a strategically selected group of major emerging economies, not as

universally generalizable results for all Global South countries. Second, the empirical model focuses on a limited set of key structural determinants due to data availability and the need to maintain a balanced panel. Key factors such as energy prices, REN policy instruments, financial development, fossil fuel subsidies, and energy infrastructure are not included in the baseline specification. While the use of second-generation panel estimators and CS-ARDL robustness checks helps mitigate potential estimation bias, concerns about missing variables cannot be eliminated. Third, potential endogeneity may arise from reverse causality, omitted variables, and unobserved common shocks. For example, GRI and RQ may support renewable ENT, but progress in REN deployment may also stimulate further innovation and institutional reforms. Although the estimators used in this study help mitigate these concerns, the findings should be interpreted as robust long-run associations rather than definitive causal evidence. Future research may extend this framework by using larger country samples, alternative institutional indicators, richer energy policy variables, and identification strategies that allow stronger causal inference.

CRediT authorship contribution statement

Abdullah Emre Caglar: Writing – review & editing, Writing – original draft, Conceptualization. **Mustafa Naimoğlu:** Writing – original draft, Visualization, Formal analysis. **Ismail Demirdag:** Writing – review & editing, Writing – original draft, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Abozeid, H. O., Elamer, A. A., & Attia, E. F. (2025). Institutional quality and sustainable firm growth: Evidence from North African countries. *Sustainable Development*, 33(3), 4380–4392. <https://doi.org/10.1002/sd.3339>
- Akorli, C. D., & Adom, P. K. (2023). The role of corruption control and regulatory quality in energy efficiency transition tendencies in Africa. *iScience*, 26, Article 106262. <https://doi.org/10.1016/j.isci.2023.106262>
- Ambec, S., Cohen, M. A., Elgie, S., & Lanoie, P. (2013). The Porter hypothesis at 20: Can environmental regulation enhance innovation and competitiveness? *Review of Environmental Economics and Policy*, 7(1), 2–22. <https://doi.org/10.1093/reep/res016>
- Anto, D., Mehta, A., & Murti, A. (2024). Using energy as a community currency for the sustainable development of newly electrified communities. *Energy for Sustainable Development*, 83, Article 101558. <https://doi.org/10.1016/j.esd.2024.101558>
- Bai, J., Kao, C., & Ng, S. (2009). Panel cointegration with global stochastic trends. *Journal of Econometrics*, 149(1), 82–99. <https://doi.org/10.1016/j.jeconom.2008.10.012>
- Bai, L., Guo, T., Xu, W., Liu, Y., Kuang, M., & Jiang, L. (2023). Effects of digital economy on carbon emission intensity in Chinese cities: A life-cycle theory and the application of non-linear spatial panel smooth transition threshold model. *Energy Policy*, 183, Article 113792. <https://doi.org/10.1016/j.enpol.2023.113792>
- Breusch, T. S., & Pagan, A. R. (1980). The Lagrange multiplier test and its applications to model specification in econometrics. *The Review of Economic Studies*, 47(1), 239–253. <https://www.jstor.org/stable/pdf/2297111>
- Caglar, A. E., Demirdag, I., Syed, Q. R., & Destek, M. A. (2026). Exploring the drivers of ecological sustainability: Insights from the world's top SDGs Performers. *Sustainable Development*, 1–15. <https://doi.org/10.1002/sd.71005>

- Caglar, A. E., Ulug, M., & Demirdag, I. (2026). The role of business R&D in environmental sustainability: Evidence from the Nordic Countries. *Corporate Social Responsibility and Environmental Management*, 1–18. <https://doi.org/10.1002/csr.70614>
- Caldara, D., & Iacoviello, M. (2022). Measuring Geopolitical Risk. *American Economic Review*, 112(4), 1194–1225. <https://doi.org/10.1257/aer.20191823>
- Çamkaya, S., Kaya, Y., & Karabayır, M. E. (2025). Do renewable and nuclear R&D expenditures affect environmental quality in France? An assessment from the perspective of the LCC hypothesis and SDGs. *Energy*, 320, Article 135179. <https://doi.org/10.1016/j.energy.2025.135179>
- Cheng, M., Wang, J., Yang, S., & Li, Q. (2024). The driving effect of technological innovation on green development: From the perspective of efficiency. *Energy Policy*, 188, Article 114089. <https://doi.org/10.1016/j.enpol.2024.114089>
- Chudik, A., & Pesaran, M. H. (2015). Common correlated effects estimation of heterogeneous dynamic panel data models with weakly exogenous regressors. *Journal of Econometrics*, 188(2), 393–420. <https://doi.org/10.1016/j.jeconom.2015.03.007>
- Depraeter, L., & Goutte, S. (2025). Geopolitical risk and clean energy investments: Exploring the role of rare earths. *International Review of Financial Analysis*, 100, Article 103965. <https://doi.org/10.1016/j.irfa.2025.103965>
- Dong, K., Zhao, J., & Taghizadeh-Hesary, F. (2023). Toward China's green growth through boosting energy transition: the role of energy efficiency. *Energy Efficiency*, 16(5), 43. <https://doi.org/10.1007/s12053-023-10123-7>
- Dumitrescu, E. I., & Hurlin, C. (2012). Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29(4), 1450–1460. <https://doi.org/10.1016/j.econmod.2012.02.014>
- Erdaş, M. L., Caglar, A. E., Uche, E., & Ahmed, Z. (2025). Future pathways for net zero emission: Analyzing the nexus of entrepreneurship and ecological sustainability development in developed economies. *Business Strategy and the Environment*, 34(5), 5644–5656. <https://doi.org/10.1002/bse.4264>
- Hartvig, A. D., Simo, M., Fazekas, D., Fragkos, P., Fragkiadakis, K., Fragkiadakis, D., & Vrontisi, Z. (2025). Comparing economic impacts of net-zero transition with climate damage uncertainties using a dual-model approach. *Environmental Research: Energy*, 2(3), Article 035005. <https://doi.org/10.1088/2753-3751/ade15e>
- He, Z. W., Lee, C. C., & Sharma, S. S. (2025). The impact of geopolitical risks on the renewable energy transition. *Energy Economics*, 143, Article 108278. <https://doi.org/10.1016/j.eneco.2025.108278>
- Herman, K. S., & Xiang, J. (2022). Channeled through trade: how foreign environmental regulations induce domestic renewable energy innovation. *Energy Research & Social Science*, 89, Article 102629. <https://doi.org/10.1016/j.erss.2022.102629>
- Hille, E. (2023). Europe's energy crisis: Are geopolitical risks in source countries of fossil fuels accelerating the transition to renewable energy? *Energy Economics*, 127, Article 107061. <https://doi.org/10.1016/j.eneco.2023.107061>
- International Energy Agency (IEA). (2025). Data and statistics. <http://www.iea.org/>
- Jaffe, A. B., & Palmer, K. (1997). Environmental regulation and innovation: A panel data study. *Review of Economics and Statistics*, 79(4), 610–619. <https://www.jstor.org/stable/2951413>
- Józwik, B., Sarigüth, S. S., Topcu, B. A., Çetin, M., & Doğan, M. (2025). Trade openness, economic growth, capital, and financial globalization: Unveiling their impact on renewable energy consumption. *Energies*, 18(5), 1244. <https://doi.org/10.3390/en18051244>
- Lazaro, L. B., Usuriaga-Najera, O. C., Neto, A. H., Grimoni, J. A. B., & Jacobi, P. R. (2025). Climate commitments and energy transition pledges in Latin America: Where is the region headed? *Energy for Sustainable Development*, 88, Article 101779. <https://doi.org/10.1016/j.esd.2025.101779>
- Le, T. H., Mai, M. L., Nguyen, M. Q. T., Nguyen, A. P., & Do, C. P. (2025). Examining the role of environmental tax, green innovation, and digital financial inclusion for energy transition: Evidence from OECD countries. *Research in Economics*, 79, Article 101034. <https://doi.org/10.1016/j.rie.2025.101034>
- Lee, C.-C., & Lee, C.-C. (2024). Not all are alike: Assessing the effect of geopolitical risk on regional renewable energy development in China. *Renewable Energy*, 222, Article 119763. <https://doi.org/10.1016/j.renene.2023.119763>
- Lee, C.-C., Wang, F., & Chang, Y.-F. (2023). Does green finance promote renewable energy? Evidence from China. *Resources Policy*, 82, Article 103439. <https://doi.org/10.1016/j.resourpol.2023.103439>
- Li, J., Zhang, X., Ali, S., & Khan, Z. (2020). Eco-innovation and energy productivity: new determinants of renewable energy consumption. *Journal of Environmental Management*, 271, Article 111028. <https://doi.org/10.1016/j.jenvman.2020.111028>
- Li, Y., Lin, Y., & Su, B. (2025). Analysis of China's energy consumption and intensity during the 13th five-year plan period. *Energy Policy*, 198, Article 114433. <https://doi.org/10.1016/j.enpol.2024.114433>
- Lopez, I., Sanagustin-Fons, V., & Fierro, J. A. M. (2026). Social perception of renewable energies: Barriers and opportunities for an inclusive energy transition. *Energy Policy*, 208, Article 114899. <https://doi.org/10.1016/j.enpol.2025.114899>
- Maghyereh, A., Boulanuouar, Z., & Essid, L. (2025). The dynamics of green innovation and environmental policy stringency in energy transition investments. *Journal of Cleaner Production*, 487, Article 144649. <https://doi.org/10.1016/j.jclepro.2024.144649>
- Martinot, E. (1998). Energy efficiency and renewable energy in Russia: Transaction barriers, market intermediation, and capacity building. *Energy Policy*, 26(11), 905–915. [https://doi.org/10.1016/S0301-4215\(98\)00022-6](https://doi.org/10.1016/S0301-4215(98)00022-6)
- McCauley, D., & Pettigrew, K. (2023). Building a just transition in asia-pacific: Four strategies for reducing fossil fuel dependence and investing in clean energy. *Energy Policy*, 183, Article 113808. <https://doi.org/10.1016/j.enpol.2023.113808>
- Muhammad, S., & Hoffmann, C. (2024). From investment to impact: The role of green finance and technological innovation on German energy transition. *Renewable Energy*, 237, Article 121665. <https://doi.org/10.1016/j.renene.2024.121665>
- Oduro, P., Uzougbo, N. S., & Ugwu, M. C. (2024). Renewable energy expansion: Legal strategies for overcoming regulatory barriers and promoting innovation. *International Journal of Applied Research in Social Sciences*, 6(5), 927–944. <https://doi.org/10.51594/ijarss.v6i5.1145>
- OECD. (2025). [https://data-explorer.oecd.org/vis?fs\[0\]=Topic%2C1%7CEnvironment%23ENV%23%7CEnvironmental%20policy%23ENV_POL%23&pg=0&fc=Topic&bp=true&snb=17&vw=tb&df\[ds\]=dsDisseminateFinalDMZ&df\[id\]=DSD_PAT_DEV%40DF_PAT_DEV&df\[ag\]=OECD.ENV.EPI&df\[vs\]=1.0&pd=1990%2C2021&dq=IDN%2BBRA%2BTHA%2BIND%2BCHN.A.ENV_PAT.ONE.T&to\[TIME_PERIOD\]=false](https://data-explorer.oecd.org/vis?fs[0]=Topic%2C1%7CEnvironment%23ENV%23%7CEnvironmental%20policy%23ENV_POL%23&pg=0&fc=Topic&bp=true&snb=17&vw=tb&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PAT_DEV%40DF_PAT_DEV&df[ag]=OECD.ENV.EPI&df[vs]=1.0&pd=1990%2C2021&dq=IDN%2BBRA%2BTHA%2BIND%2BCHN.A.ENV_PAT.ONE.T&to[TIME_PERIOD]=false) (Accessed: 23.08.2025).
- Olaniyi, C. O., Al-Faryan, M. A. S., & Ogbaro, E. O. (2025). Do institutional quality and its threshold matter in the sensitivity of the renewable energy transition to financial development? New empirical perspectives. *International Journal of Finance and Economics*, 30, 5–43. <https://doi.org/10.1002/ijfe.2900>
- Our World in Data (OWD). (2025). Energy database. <https://ourworldindata.org/search?q=renewable+fossil> (Accessed: 20.08.2025).
- Özbek, S., & Naimoğlu, M. (2025). The effectiveness of renewable energy technology under the EKC hypothesis and the impact of fossil and nuclear energy investments on the UK's ecological footprint. *Energy*, 322, Article 135351. <https://doi.org/10.1016/j.energy.2025.135351>
- Pata, U. K., Alola, A. A., Erdogan, S., & Kartal, M. T. (2023). The influence of income, economic policy uncertainty, geopolitical risk, and urbanization on renewable energy investments in G7 countries. *Energy Economics*, 128, Article 107172. <https://doi.org/10.1016/j.eneco.2023.107172>
- Pesaran, M. H. (2004). General diagnostic tests for cross section dependence in panels. *IZA Discussion Paper*, 1240, 1–39. <https://repec.iza.org/dp1240>
- Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265–312. <https://doi.org/10.1002/jae.951>
- Pesaran, M. H., & Yamagata, T. (2008). Testing slope homogeneity in large panels. *Journal of Econometrics*, 142(1), 50–93. <https://doi.org/10.1016/j.jeconom.2007.05.010>
- Porter, M. E. (1991). Essay. *Scientific American*, 264(4), 168. <https://www.jstor.org/stable/2486436>
- Porter, M. E., & van der Linde, C. (1995). Toward a new conception of the environment–competitiveness relationship. *Journal of Economic Perspectives*, 9(4), 97–118. <https://www.jstor.org/stable/2138392>
- Shahzad, U. (2025). Achieving clean energy transitions: How green innovation and financial development shape energy usage. *Energy Economics*, 150, Article 108851. <https://doi.org/10.1016/j.eneco.2025.108851>
- Shittu, I., Saqib, A., Tang, C. F., & Chen, Y. (2025). Does global geopolitical risk jeopardize the clean energy transition? New evidence from a global panel. *Renewable and Sustainable Energy Reviews*, 223, Article 116006. <https://doi.org/10.1016/j.rser.2025.116006>
- Tiwari, S., Shahzad, U., Alofaysan, H., Walsh, S. T., & Kumari, P. (2024). How do renewable energy, energy innovation and climate change shape the energy transition in USA? Unraveling the role of green finance development. *Energy Economics*, 140, Article 107947. <https://doi.org/10.1016/j.eneco.2024.107947>
- Uzar, U. (2020). Political economy of renewable energy: Does institutional quality make a difference in renewable energy consumption? *Renewable Energy*, 155, 591e603. <https://doi.org/10.1016/j.renene.2020.03.172>
- Wang, H., Zhang, T., Zhang, Z., & Feng, Y. (2025). What is the global causality between geopolitical risks, government governance, and energy transition? Empirical evidence from cross-country data. *Carbon Balance and Management*, 20, 31. <https://doi.org/10.1186/s13021-025-00322-3>
- Wang, S., Wang, J., & Wang, W. (2023). Do geopolitical risks facilitate the global energy transition? Evidence from 39 countries in the world. *Resources Policy*, 85, Article 103952. <https://doi.org/10.1016/j.resourpol.2023.103952>
- Wei, J., Wen, J., Wang, X.-Y., et al. (2023). Green innovation, natural extreme events, and energy transition: Evidence from Asia-Pacific economies. *Energy Economics*, 121. <https://doi.org/10.1016/j.eneco.2023.106638>
- Westerlund, J. (2008). Panel cointegration tests of the Fisher effect. *Journal of Applied Econometrics*, 23(2), 193–233. <https://doi.org/10.1002/jae.967>
- World Development Indicators (WDI). (2025). <https://databank.worldbank.org/source/world-development-indicators> (Accessed: 21.09.2025).
- Worldwide Governance Indicators (WGI). (2025). <https://www.worldbank.org/en/publication/worldwide-governance-indicators> (Accessed: 21.09.2025).
- Xu, R., Chen, X., & Dong, P. (2024). Nexus among financial technologies, oil rents, governance and energy transition: Panel investigation from Asian Economies. *Resources Policy*, 90, Article 104746. <https://doi.org/10.1016/j.resourpol.2024.104746>
- Zhang, J., & Usman, M. (2025). Redefining energy policy for sustainable growth: The interplay of fossil fuel subsidies, energy security risks, and energy balances in shaping geopolitical stability. *Energy*, 322, Article 135620. <https://doi.org/10.1016/j.energy.2025.135620>
- Zhang, Y., Alharthi, M., Ali, S. A., Abbas, O., & Taghizadeh-Hesary, F. (2022). The eco-innovative technologies, human capital, and energy pricing: Evidence of sustainable energy transition in developed economies. *Applied Energy*, 325, Article 119729. <https://doi.org/10.1016/j.apenergy.2022.119729>
- Zhao, D., e-Ali, M. S., Chaudhry, M. O., Ayub, B., Waqas, M., & Ullah, I. (2024). Modeling the nexus between geopolitical risk, oil price volatility and renewable energy investment: Evidence from Chinese listed firms. *Renewable Energy*, 225, Article 120309. <https://doi.org/10.1016/j.renene.2024.120309>
- Zhu, Z., Hunjra, A. I., Alharbi, S. S., & Zhao, S. (2025). Global energy transition under geopolitical risks: An empirical investigation. *Energy Economics*, 145, Article 108495. <https://doi.org/10.1016/j.eneco.2025.108495>